

HantaBERT: Multi-Task Hantavirus Classification with DNABERT-2 Fine-Tuning

Muhammad Faiz Atharrahman¹
18222063

Muhammad Rafi Dhiyaulhaq²
18222069

Lydia Gracia³
18222035

Information Systems and Technology Major
School of Electrical Engineering and Informatics
Institut Teknologi Bandung

¹m@faiz.at · ²rafidhiyaulh@gmail.com · ³ly.gracies@gmail.com

Abstract—Hantaviruses are a group of zoonotic RNA viruses transmitted primarily by rodents and capable of causing severe hemorrhagic fever in humans. Rapid classification of a new sequence (covering the virus species, the likely reservoir host, and its geographic origin) is a crucial step in outbreak response. This study proposes *HantaBERT*, a multi-task model obtained by *fine-tuning* the genome foundation model DNABERT-2 on 8,822 hantavirus sequences from NCBI GenBank. The proposed architecture places a shared *bottleneck* on top of the DNABERT-2 *encoder*, then branches into three independent classification heads for predicting species (23 classes), host (3 classes), and geographic origin (7 continents). Training is performed with a weighted cross-task loss $L = 1.0 L_{sp} + 0.5 L_{host} + 0.3 L_{geo}$, a differential encoder-vs-head learning rate, and memory optimization via *gradient accumulation* + AMP fp16. On a held-out test set, the model achieves 96.7% accuracy (species), 91.4% (host), and 80.5% (geography). A UMAP visualization of the *bottleneck embedding* reveals separation of viral lineages as well as substructure per genome segment (*S/M/L*) without any explicit *clustering* objective, indicating the emergence of a meaningful phylogenetic representation. The code is available at <https://github.com/HantaBERT>.

Index Terms—hantavirus, DNABERT-2, multi-task learning, nucleotide sequence classification, genome embedding, UMAP

I. INTRODUCTION

A. Hantavirus Biology Concepts

Hantaviruses (family *Hantaviridae*, genus *Orthohantavirus*) are negative-sense single-stranded RNA viruses with a tripartite segmented genome: the **S** segment (*small*, ~1.8 kb) encodes the nucleocapsid (N) protein that wraps the genome; the **M** segment (*medium*, ~3.6 kb) encodes a glycoprotein precursor that is cleaved into the envelope-forming Gn and Gc; and the **L** segment (*large*, ~6.5 kb) encodes the RNA-dependent RNA polymerase (RdRp). These viruses are transmitted primarily by rodents, with several species able to cause two severe clinical syndromes in humans: *Hantavirus Cardiopulmonary Syndrome* (HCPS) in the Americas and *Hemorrhagic Fever with Renal Syndrome* (HFRS) in Eurasia. Rapid identification of new sequences is an important component of outbreak response and epidemiological surveillance.

B. Related Computational Methods

Classical approaches based on *alignment* and phylogenetic trees (*neighbor-joining*, *maximum likelihood*) yield accurate

results but require compute-intensive preprocessing and are sensitive to the choice of reference region. Deep learning approaches offer an alternative path: *Convolutional Neural Networks* (CNNs) for viral taxonomic classification and, more recently, Transformer foundation models for nucleotide sequences. DNABERT [3] trains a BERT *encoder* [4] on the human genome with *k-mer* tokenization; DNABERT-2 [1] extends it to multiple species with a more efficient *Byte-Pair Encoding* tokenization; DNABERT-S [2] adds a *species-aware* objective at the pre-training stage. *Multi-task learning* itself has long been recognized as a form of implicit regularization [5], and UMAP [6] is widely used to nonlinearly visualize high-dimensional *embeddings*.

C. Research Questions

This study sets out from three questions:

- RQ1** To what extent can multi-task *fine-tuning* of DNABERT-2 with a single shared *bottleneck* produce competitive accuracy across three hantavirus attributes (species, host, geography) simultaneously?
- RQ2** Does the *embedding* output of the *bottleneck* reflect the phylogenetic structure of the viruses, even though the model is never trained with an explicit *clustering* objective?
- RQ3** What biological properties emerge from the analysis of intra-species substructure in that *embedding*?

D. Contributions

The contributions of this study are summarized as follows:

- Proposing *HantaBERT*: a multi-task architecture that wraps DNABERT-2 with a shared *bottleneck* and three independent classification heads for species, host, and geography.
- Designing a weighted multi-task loss with *balanced* class weights to handle the extremely skewed distribution of hantavirus lineages.
- Reporting 96.7%/91.4%/80.5% accuracy on a held-out test set and showing that the *bottleneck embedding* spontaneously forms lineage clusters *plus* substructure per genome segment (*S/M/L*).

Table I
DATASET COMPOSITION AFTER PREPROCESSING.

Subset	N seq.	Task	Classes	Weight
Train	7,057	Species	23	<i>balanced</i>
Val	882	Host	3	<i>balanced</i>
Test	883	Geography	7	<i>balanced</i>
Total	8,822			

- Open-sourcing the code, hyperparameters, and evaluation scripts at <https://github.com/HantaBERT> for reproduction and further experimentation.
- Developing a public web interface based on FastAPI + static HTML/JS that wraps the ready-to-use model for interactive inference: input a DNA/RNA sequence or a FASTA file, output probabilistic predictions per task with *top-N ranking* and a geographic map visualization.

II. METHODS

A. Dataset and Preprocessing

The base dataset was assembled from hantavirus submissions on NCBI GenBank and processed through an internal pipeline. Each row contains a nucleotide sequence, a segment label (*S/M/L*), a species label, a host label (*Rodent/Human/Others/Unknown*), and geographic coordinates when available. Sequences are stored in DNA representation (*T*, not *U*) so they can be passed directly to the DNABERT-2 tokenizer.

The source *geo_label_broad* column showed extensive incompleteness and was repaired through batch *reverse geocoding* of latitude/longitude pairs. Mapping country codes to continents using the *CC_TO_CONTINENT* table yields seven classes (Americas, Europe, Asia, Africa, Oceania, Other, *Unknown*). Species with < 30 samples were merged into an *Other* class, *Orthohantavirus sp.* entries were dropped, and 579 *Unknown*-host rows were removed. The data was split with stratification on the species label (80/10/10) into 7,057/882/883 sequences (Table I). Class weights were computed with the *balanced* strategy from *sklearn* for each task.

B. HantaBERT Architecture

HantaBERT consists of three main components: a pre-trained DNABERT-2 *encoder*, one shared *bottleneck* layer, and three independent classification heads. Its computation flow is shown in Figure 1. Given an input nucleotide sequence of at most $L_{\max} = 512$ BPE tokens, the computation flow is:

$$\begin{aligned}
\mathbf{H} &= \text{DNABERT2}(\mathbf{x}) \in \mathbb{R}^{L \times 768}, \\
\mathbf{h} &= \text{MeanPool}(\mathbf{H}, \mathbf{m}) \in \mathbb{R}^{768}, \\
\mathbf{z} &= \text{Dropout}(\text{GELU}(\text{LayerNorm}(W\mathbf{h} + b))), \\
\mathbf{y}_{\text{task}} &= W_{\text{task}} \text{Dropout}(\mathbf{z}) + b_{\text{task}}, \quad \text{task} \in \{\text{sp}, \text{host}, \text{geo}\}.
\end{aligned}$$

Mean pooling is computed as the *attention mask*-weighted mean $\mathbf{h} = \sum_i m_i \mathbf{H}_i / \max(\sum_i m_i, \epsilon)$ so that *padding* tokens are not taken into account. The *bottleneck* $W \in \mathbb{R}^{768 \times 768}$ is

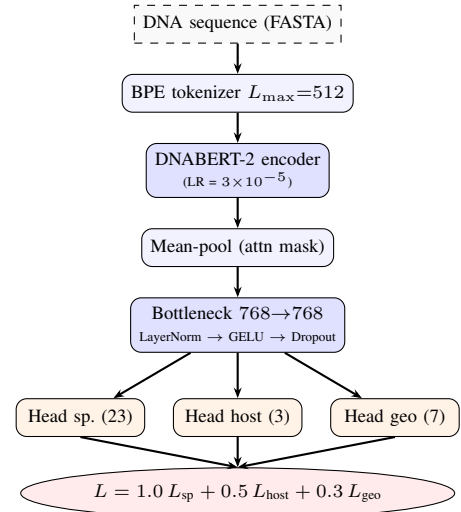


Figure 1. HantaBERT computation flow. Dark-blue components are trained with the encoder *learning rate* (3×10^{-5}); light-blue and orange components with an LR ten times larger.

task-agnostic: its parameters are trained by all three tasks at once, forcing the representation $\mathbf{z}$ to carry information useful for every task. Each head has an independent *dropout* of 0.1 before the final linear layer to dampen cross-task gradient interference.

C. Multi-Task Loss Function

The overall *loss* is a linear combination of three class-weighted *cross-entropy* terms:

$$L = \lambda_{\text{sp}} L_{\text{sp}} + \lambda_{\text{host}} L_{\text{host}} + \lambda_{\text{geo}} L_{\text{geo}},$$

with $\lambda_{\text{sp}}=1.0$, $\lambda_{\text{host}}=0.5$, and $\lambda_{\text{geo}}=0.3$. The weights reflect relative priority: species is the primary task with the cleanest supervision, while geography has the lowest signal-to-noise ratio.

D. Training Strategy

The training configuration is summarized in Table II. The pre-trained *encoder* is given a learning rate ten times smaller than the new layers (*bottleneck* and heads). Because the 512-token *full attention* of DNABERT-2 cannot fit a *batch* of 16 on a *consumer-grade* GPU, an effective *batch size* of 16 is reached via *BATCH_SIZE*=4 and *GRAD_ACCUM_STEPS*=4. In addition, the *attention_probs_dropout_prob* parameter is set to 0.1 to force the standard PyTorch *attention* path, avoiding the DNABERT-2 Triton *kernel* that exceeds the *shared memory* limit.

E. Web Interface for Inference

To make the model ready for use by non-programmer researchers, a public web interface was developed. The *backend* (HantaBERT-API) wraps the *MultiTaskHantaBERT checkpoint* behind a FastAPI + Uvicorn service, packaged as a Docker *container* based on *python:3.11-slim*,

Table II
MAIN TRAINING HYPERPARAMETERS.

Parameter	Value
Backbone	zhihan1996/DNABERT-2-117M
Max token length	512 BPE
Batch size	4 ($\times 4$ grad accum = effective 16)
LR encoder / heads	3×10^{-5} / 3×10^{-4}
Optimizer	AdamW, $\beta = (0.9, 0.999)$, wd = 0.01
Warmup / total epochs	200 linear steps / 10
Mixed precision	AMP fp16
Grad clipping	1.0 (global norm)
$\lambda_{sp}/host/geo$	1.0/0.5/0.3

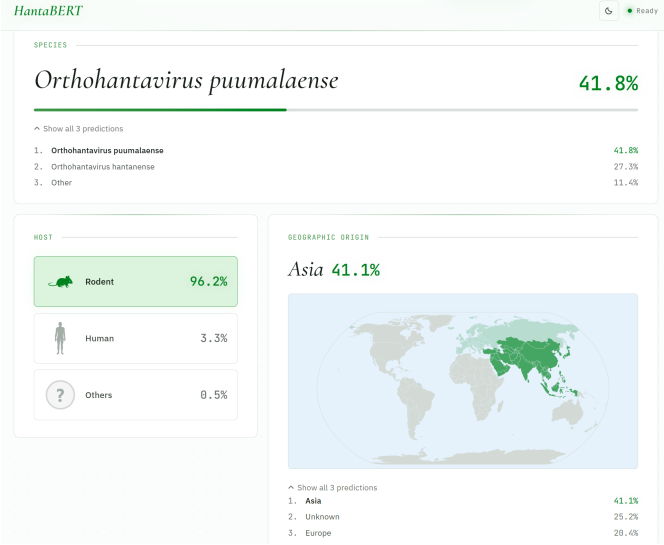


Figure 2. Screenshot of the HantaBERT web interface for a 224 bp test sequence ATGAAAGACCTTCTGAAG...GAGAAGCTGGAG. The *top-1* predictions: *Orthohantavirus puumalaense* (41.8%), host *Rodent* (96.2%), and geographic region Asia (41.1%). The *top-3* probabilities can be expanded on each task card; the geography result is visualized on an interactive world map.

and deployed at hantabert-api.faizath.com. The frontend (HantaBERT-Web) is a set of pure static HTML/CSS/JavaScript files with no JS framework; the interactive world map for geographic origin is drawn using the D3 and TopoJSON libraries loaded from a CDN.

Users can paste a raw DNA/RNA sequence or upload a FASTA file; the *U* bases in RNA input are automatically converted to *T* before BPE tokenization, and the system displays a warning when a sequence exceeds 512 tokens and is therefore truncated. The number of *top-N* results (1 to 23) can be configured per task, complete with an expandable confidence score. Figure 2 shows the system output for a 224 bp test sequence run directly on the interface.

F. Code and Data Availability

All artifacts are released openly under the GitHub organization <https://github.com/HantaBERT>, comprising three repositories: (i) HantaBERT: the *preprocess*, *train*, *evaluate*, *visualize* code, hyperparameters, and model *check-*

Table III
TRAINING PROGRESS ON THE VALIDATION SET (10 EPOCHS).

Ep	L_{tr}	L_{val}	Sp.	Host	Geo
1	3.024	2.443	78.9%	72.8%	72.3%
3	1.034	1.195	92.9%	83.1%	74.1%
5	0.634	1.150	94.0%	87.1%	77.6%
7	0.452	0.869	96.5%	90.4%	78.1%
9	0.336	0.818	96.3%	91.4%	80.3%
10	0.307	0.797	96.7%	91.4%	80.5%

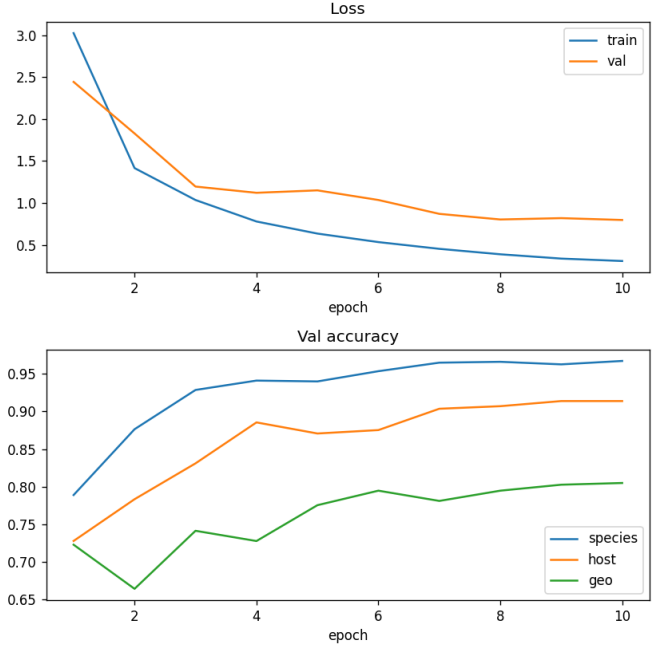


Figure 3. Loss curves (top) and per-task validation accuracy (bottom) over 10 epochs.

point; (ii) HantaBERT-API: the FastAPI inference service together with its *Dockerfile*; (iii) HantaBERT-Web: the static frontend files. Each repository includes a *requirements.txt*/README listing the tested package versions and a step-by-step reproduction guide. In addition, a link to a recorded presentation video is provided at <https://youtu.be/4Ia0FM7iYb4> and the presentation slides can be accessed at <https://canva.link/33nwtodm1si8nxw>.

III. RESULTS AND DISCUSSION

A. Training Progress

Table III shows the progress at each epoch on the validation set. After 10 epochs, *val_loss* dropped from 2.443 to 0.797 and validation accuracy reached 96.7% (species), 91.4% (host), 80.5% (geography). The species task saturated the fastest, consistent with its cleanest supervision and a class count that makes the discriminative signal rich. The curves in Figure 3 show the training-validation *loss* gap shrinking over epochs with no clear sign of overfitting.

Table IV
ACCURACY ON THE TEST SET.

Task	Neural head
Species (23)	96.7%
Host (3)	91.4%
Geography (7)	80.5%

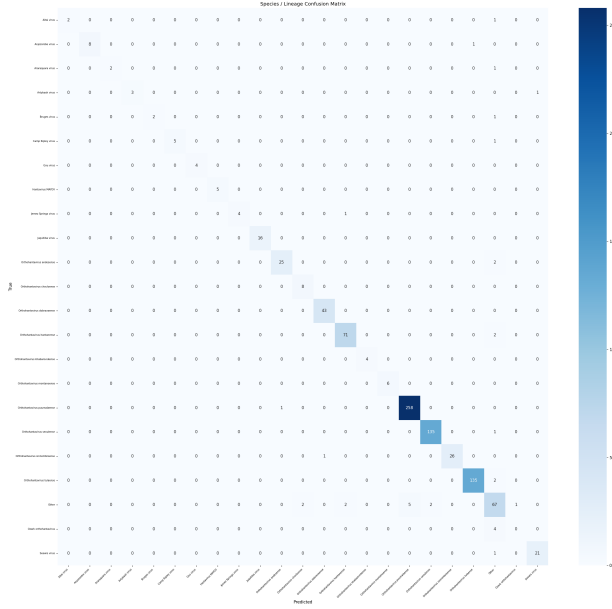


Figure 4. Confusion matrix for species classification on the test set.

B. Evaluation on the Test Set

Table IV summarizes the test-set accuracy for two evaluation paths: the post-training neural heads and an SVM-RBF baseline ($C = 10$, *balanced* classes) over the frozen and standardized *bottleneck embedding*. The purpose of this baseline is to measure the quality of the *embedding* in isolation (not to beat the neural heads) and to answer **RQ2**: if a simple SVM-RBF can achieve high accuracy over **z**, it means the *bottleneck* has indeed mapped the sequences into a space whose classes are non-linearly separable.

C. Error Analysis and Biological Context

Figure 4 shows the *confusion matrix* for the species task; dominant *lineages* (e.g. *O. seoulense*, *O. puumalaense*) are predicted with high accuracy, and the errors concentrate on the *Other* class, which is an aggregate of rare *lineages*, a direct consequence of the <30 -sample threshold.

In the host task, most errors are confusions between *Rodent* and *Others*, which is reasonable because some hantaviruses (e.g. *Thottimvirus*) are also isolated from non-rodent *Insectivora* (bats, shrews). In the geography task, errors concentrate on *cosmopolitan* *lineages*: the Seoul virus carried by *Rattus norvegicus* is found on every continent following the distribution of its host rat, so the continent label is not always deterministic from the sequence; the 80.5% accuracy is likely an upper bound close to the inherent ambiguity of that label.

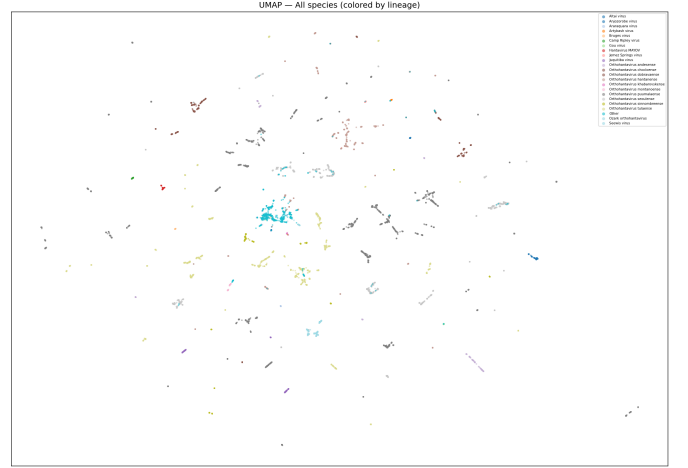


Figure 5. UMAP projection of 8,822 sequences colored by lineage.

D. UMAP Visualization and Phylogenetic Substructure

To answer **RQ2** and **RQ3**, UMAP ($n_{\text{neighbors}}=20$, $\text{min_dist}=0.1$, *cosine* metric) was applied to the *bottleneck embedding* of all sequences; a variant with $n_{\text{neighbors}}=15$, $\text{min_dist}=0.05$ was used for the intra-species plots. Figure 5 shows the *embedding* forming separate clusters per *lineage*, a result that is not explicitly supervised by any *clustering* objective.

The intra-species plots (Figures 6 and 7) reveal an interesting phenomenon: the *embedding* groups sequences by **genome segment** (*S/M/L*) even though the segment is never a supervision target. This is consistent with hantavirus biology: the three segments encode proteins with different selective pressures and nucleotide content: N (segment S) is highly *conserved*, Gn/Gc (segment M) undergo *positive selection* at envelope antigenic sites, and RdRp (segment L) preserves the active polymerase motifs. These differences in *di-* and *tri-nucleotide* composition are detected by the DNABERT-2 BPE *tokenizer* and manifest in the *embedding* as cluster separation, answering **RQ3**: the model has implicitly mapped the structural information of the hantavirus genome.

IV. CONCLUSION

This study presents HantaBERT, a multi-task model obtained by *fine-tuning* DNABERT-2 to classify the species, host, and geographic origin of hantavirus sequences in a single *forward pass*. The shared *bottleneck* scheme, the weighted multi-task *loss*, and the memory-friendly training strategy (AMP fp16, *grad accum*, differential LR) yield 96.7%/91.4%/80.5% accuracy on a held-out test set, answering **RQ1** affirmatively. The 768-d *embedding* that emerges from training not only forms lineage clusters (**RQ2**) but also substructure per genome segment (*S/M/L*) consistent with viral biology (**RQ3**). Objectively, this contribution provides a compact computational basis for triaging new hantavirus sequences: multi-attribute classification + a visualization-ready representation with a single model. Complementing the *engineering* contribution,

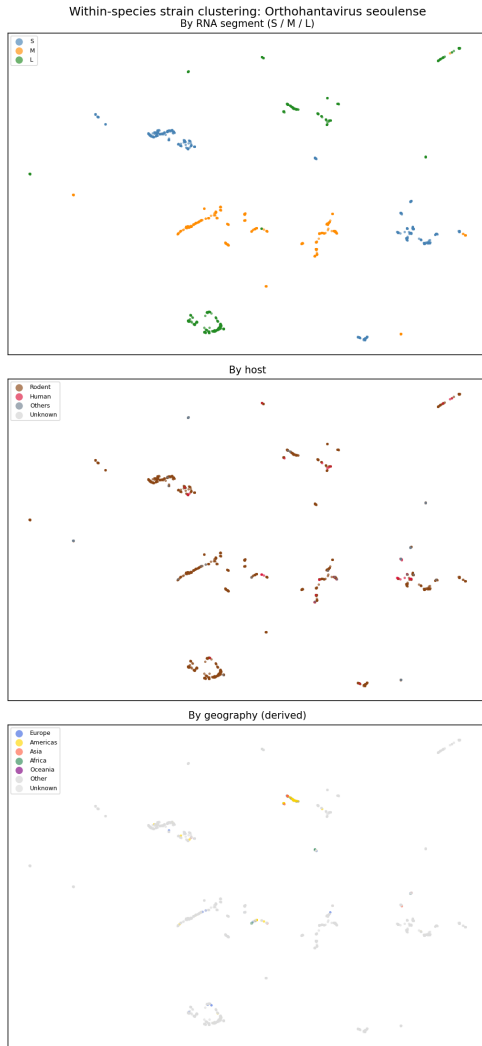


Figure 6. Intra-species UMAP of *O. seoulense* (top-to-bottom: segment / host / geography).

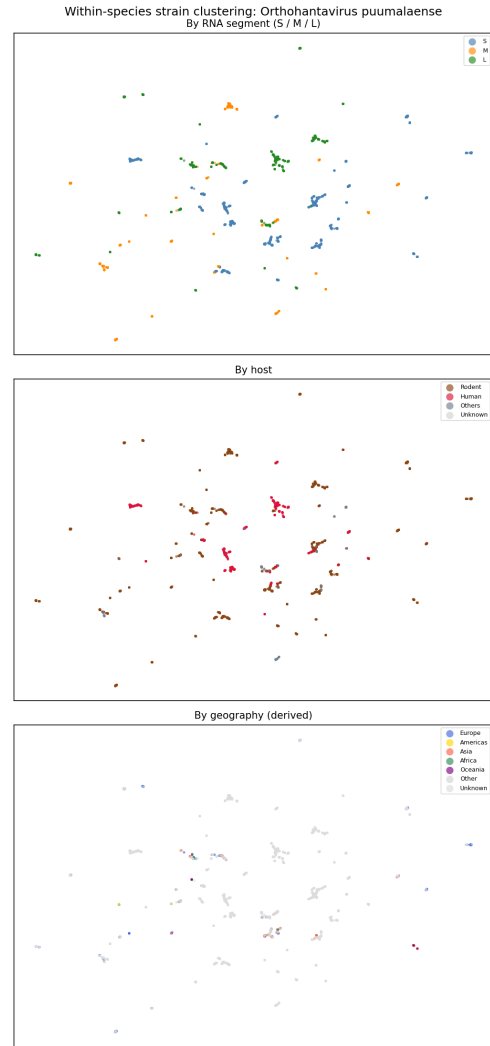


Figure 7. Intra-species UMAP of *O. puumalaense* (top-to-bottom: segment / host / geography).

the model is *deployed* behind a public web interface (FastAPI + static HTML/JS frontend) that accepts DNA/RNA sequences or FASTA files and returns probabilistic predictions per task together with an interactive geographic map.

Directions for future work: (i) extending the context beyond 512 tokens so that long L segments are not truncated; (ii) a probability calibration head (*temperature/Platt scaling*) for reliable uncertainty estimation in surveillance; (iii) a cross-genome-segment *ensemble* when all three segments of one isolate are available; (iv) exploring DNABERT-S as an alternative *backbone* because its pre-training objective is *species-aware*.

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